

Swarm Intelligence

CSC3034 Computational Intelligence

Dr Tang Tiong Yew

Introduction to swarm intelligence

Particle swarm optimisation

- Background

- Algorithm

- Variants

- Example

Ant colony optimisation

- Background

- Algorithm

- Example

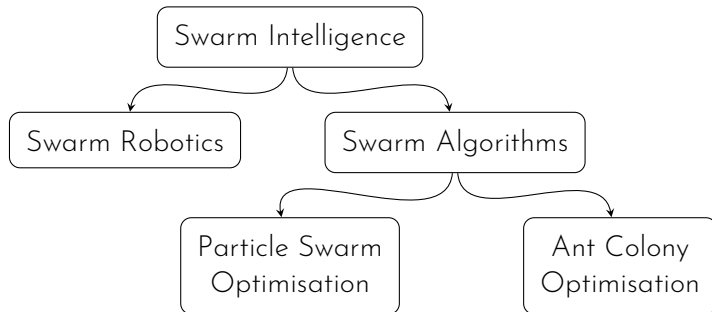
- Variants

Other swarm algorithms



Swarm intelligence is a subfield in AI focusing on the **collective** behaviour of **decentralised** and **self-organised** systems comprised of relatively **simple** agents **interacting** locally with one another and with the environment.¹

¹A. Iglesias, A. Gálvez, and P. Suárez, "Swarm robotics – A case study: Bat robotics", Nature-Inspired Computation and Swarm Intelligence, pp. 273-302, 2020.



Particle swarm optimisation (PSO)

Background

- ▶ Inspired from the flocking behaviour of birds

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- ▶ It's later used/improved to create the flocking algorithm which simulates the flocking behaviour of birds.
- ▶ Eventually the changes made by researchers led it to be used for more generic simulation than just simulating bird models.

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- ▶ particles are “flown” through the search space to find the best position.
- ▶ the direction towards which each particle “fly” is based on its own experience and the experience of the neighbours.

- ▶ parameters of a particle are its position $x_i(t)$ and velocity $v_i(t)$

Basic principles

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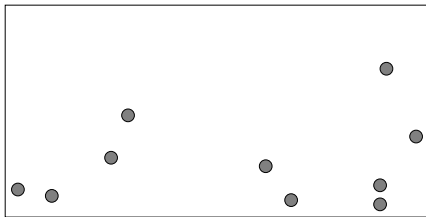
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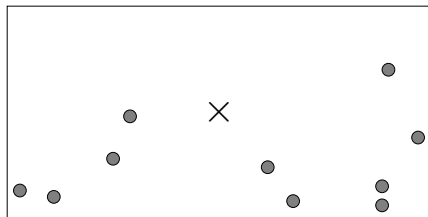


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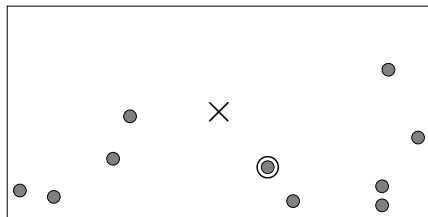


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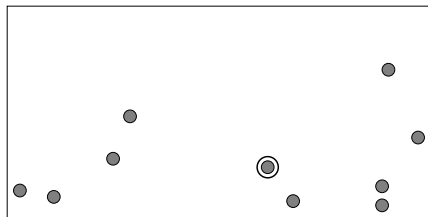


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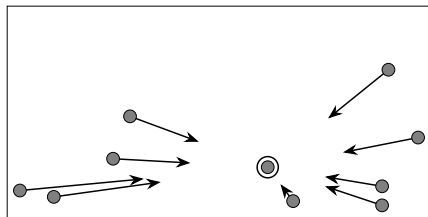


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Global best PSO

- ▶ also known as gbest PSO

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- ▶ neighbourhood for a particle is the entire swarm (population), i.e. all particles affect each other directly

The velocity of the particle is calculated as

$$v_i(t+1) = v_i(t) + \alpha_1 \beta_1(t) (p_i(t) - x_i(t)) + \alpha_2 \beta_2(t) (p_g(t) - x_i(t))$$

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Current velocity

Effect of personal
best position

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positive acceleration
constant

The velocity of the particle is calculated as

$$v_i(t+1) = v_i(t) + \alpha_1 \overset{\substack{\text{random values} \\ (0,1)}}{\beta_1(t)} (p_i(t) - x_i(t)) + \alpha_2 \beta_2(t) (p_g(t) - x_i(t))$$

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Diagram illustrating the velocity calculation formula with annotations:

- α_1 : positive acceleration constant
- $\beta_1(t)$: random values (0,1)
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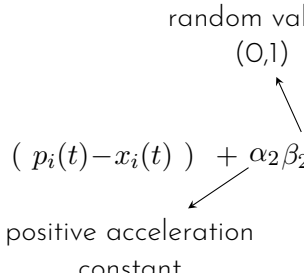


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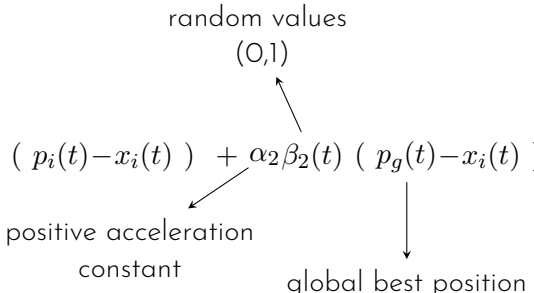
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- $\alpha_2 \beta_2(t)$: global best position
- $p_g(t) - x_i(t)$: global best position

Algorithm 1 gbest PSO

```

Create and initialise an  $n_x$ -dimensional swarm
repeat

```

until stopping condition is true

Create and initialise an n_x -dimensional swarm

for each particle do

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Create and initialise an n_x -dimensional swarm

for each particle **do**

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end for

for each particle **do**

update velocity

end for

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Algorithm 1 gbest PSO

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3. Repeat until termination
 - 3.1 loop through all particles to set personal and neighbourhood best position
 - 3.2 loop through all particles to update velocity and position

Example

Problem: Find the minimum of $f(x) = x^2$ using gbest PSO.

Local best PSO

- ▶ also known as lbest PSO

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- ▶ also known as lbest PSO
- ▶ a smaller neighbourhood is defined for each particle
- ▶ the velocity of the particle is calculated as

$$v_i(t+1) = v_i(t) + \alpha_1 \beta_1(t)(p_i(t) - x_i(t)) + \alpha_2 \beta_2(t)(p_g(t) - x_i(t))$$

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the function is the same as that in gbest PSO with the exception that $p_g(t)$ is the neighbourhood best position of the particle

$$\begin{array}{c} \text{Next velocity} \\ \uparrow \\ \underline{v_i(t+1)} = \underline{v_i(t)} + \frac{\alpha_1 \beta_1(t) (p_i(t) - x_i(t))}{\text{Effect of personal best position}} + \frac{\alpha_2 \beta_2(t) (p_g(t) - x_i(t))}{\text{Effect of neighbourhood best position}} \end{array}$$

Current velocity

Local best PSO

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Diagram illustrating the Local best PSO velocity update equation with annotations:

- $\alpha_1 \beta_1(t)$: positive acceleration constant
- $\beta_1(t)$: random values (0,1)
- $p_i(t)$: personal best position
- $x_i(t)$: current position

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Diagram illustrating the components of the velocity update equation for Local Best PSO:

- α_1 : positive acceleration constant
- $\beta_1(t)$: random values (0,1)
- $p_i(t) - x_i(t)$: current position (relative to personal best)
- α_2 : positive acceleration constant
- $\beta_2(t)$: random values (0,1)
- $p_g(t) - x_i(t)$: neighbourhood best position (relative to global best)

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Algorithm 2 lbest PSO

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Definition of neighbourhood of a particle

- ▶ based on particle indices. If a neighbourhood is defined to have the size of n_N , the neighbourhood of the particle x_i is

$$x_{i-n_N}(t), x_{i-n_N+1}(t), \dots, x_{i-1}(t), x_i(t), x_{i+1}(t), \dots, x_{i+n_N}(t)$$

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- ▶ based on spatial similarity. Neighbourhood can be defined based on the (Euclidean) distance. Particles that are within a certain proximity of a particle is included in its neighbourhood.

Global best PSO vs local best PSO

- ▶ gbest PSO converges faster than lbest PSO, but less diversity

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- ▶ gbest PSO converges faster than lbest PSO, but less diversity
- ▶ lbest PSO has larger diversity, thus less possible to be trapped in local minima.

Termination conditions

Termination criteria

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- ▶ terminate when an acceptable solution has been found
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- ▶ terminate when the particles are too close together
- ▶ terminate when the change in fitness is approximately zero

Variants of PSO are introduced to tweak different aspects of the generic PSO algorithm.

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To reduce the possibility of particles flying out of the search space, **velocity clamping** is introduced. This limits the velocity of the particle v_i to $[-v_{\max}, v_{\max}]$.

In this equation,

$$v_i(t+1) = v_i(t) + \alpha_1 \beta_1(t)(p_i(t) - x_i(t)) + \alpha_2 \beta_2(t)(p_g(t) - x_i(t))$$

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without velocity memory, $v_i(t)$,

→ the swarm converges to local best solution,

with velocity memory, $v_i(t)$,

→ the swarm expands to find global best solution.

Inertia weight, w was introduced to control/balance this behaviour

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$$v_i(t+1) = wv_i(t) + \alpha_1\beta_1(t)(p_i(t) - x_i(t)) + \alpha_2\beta_2(t)(p_g(t) - x_i(t))$$

Alternatively, **constriction factor**, χ was introduced to control this behaviour

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$$v_i(t+1) = \chi \{v_i(t) + \alpha_1 \beta_1(t)(p_i(t) - x_i(t)) + \alpha_2 \beta_2(t)(p_g(t) - x_i(t))\}$$

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$$\text{where } \chi = \frac{2}{|2 - \alpha - \sqrt{\alpha^2 - 4\alpha}|}$$

with $\alpha = \alpha_1 + \alpha_2$ and $\alpha > 4$

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with $\alpha = \alpha_1 + \alpha_2$ and $\alpha > 4$

with the introduction of constriction factor, velocity clamping is no longer needed.

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This variant of PSO only considers personal best and not group best. The velocity update equation becomes

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Social-only model

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Social-only model

This variant of PSO only considers global best and not personal best. The velocity update equation becomes

$$v_i(t+1) = v_i(t) + \alpha_2 \beta_2(t)(p_g(t) - x_i(t))$$

Variants

Selfless model

Selfless model

Selfless model

This variant of PSO is basically the social model, but with the condition that, the neighbourhood best cannot be the particle itself.

Example

Problem: Find the maximum value of $f(x) = 15x - x^2$ assuming x can only be integers between 0 and 15

1. using gbest PSO.
2. using lbest PSO.

Ant colony optimisation (ACO)

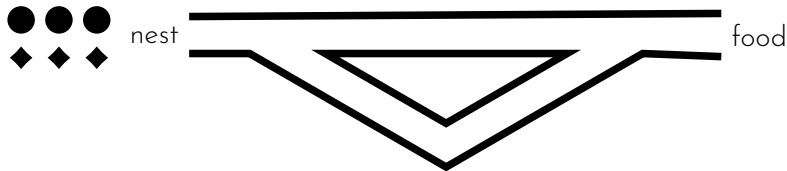
Background

- ▶ Ants are efficient at bringing food back to their nest (foraging behaviour).
- ▶ Ants left behind a chemical pheromone trail along their path. Ants also tend to follow a path with higher pheromone concentration.

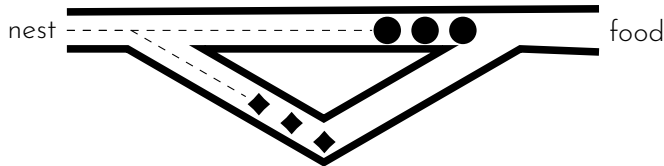
Ant colony optimisation (ACO)

Background

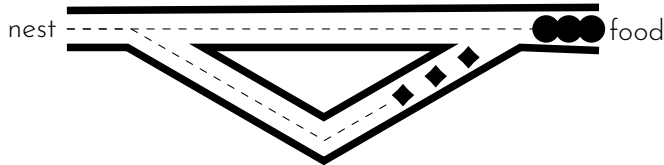
- ▶ Ants are efficient at bringing food back to their nest (foraging behaviour).
- ▶ Ants left behind a chemical pheromone trail along their path. Ants also tend to follow a path with higher pheromone concentration.



- ▶ At the initial stages of foraging, ants will take all possible routes to the food with same probability.

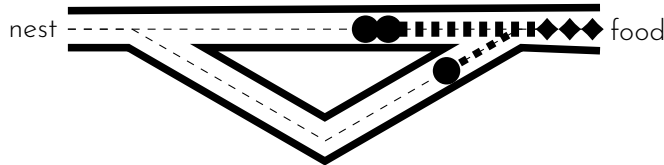


- ▶ As the ants on the shorter route will arrive at the food earlier, that's the only route with pheromone deposited, it is more likely that the ants will choose the shorter route to return to the nest.

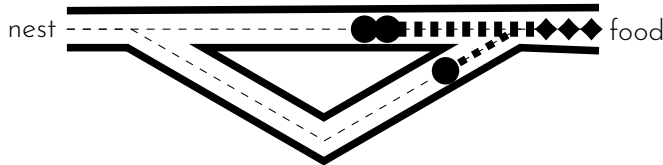


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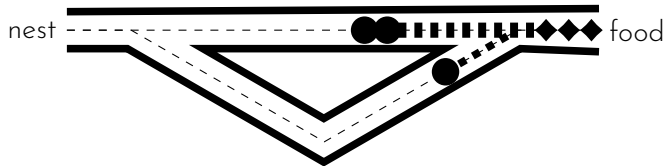


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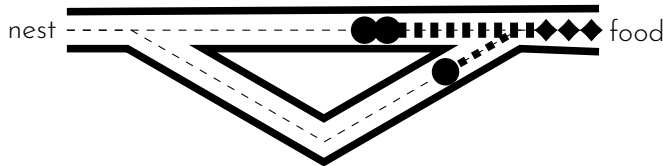
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- ▶ The pheromone trail on longer route will evaporate over time.
- ▶ Eventually all ants will take the shortest route.
- ▶ This indirect communication between ants via pheromone trails is known as stigmergy.

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- A set of small navigation icons typically found in Beamer presentations, including symbols for back, forward, search, and other slide navigation functions.

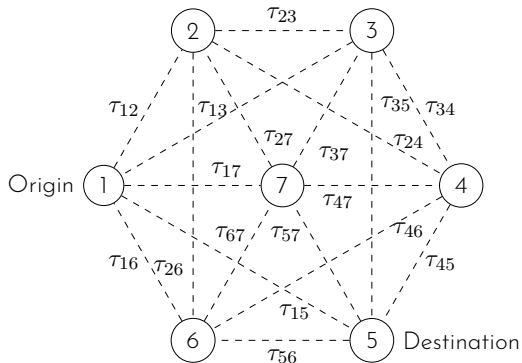
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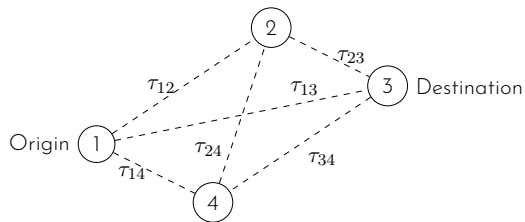
- ▶ The foraging behaviour of ants inspired the introduction of the ant colony optimisation (ACO) algorithms (or some call ant colony system (ACS)).
- ▶ ACO is used to solve combinatorial optimisation problems, i.e. to identify the best (optimal) combination (ordering) from a finite set of discrete objects.
- ▶ The classic example: the **travelling salesman problem** (TSP):
Given a list of cities and the distances between each pair of cities, what is the shortest possible route that visits each city and returns to the origin city?

- Generally, a combinatorial optimisation problem is a problem of finding the shortest path between two nodes on a graph.

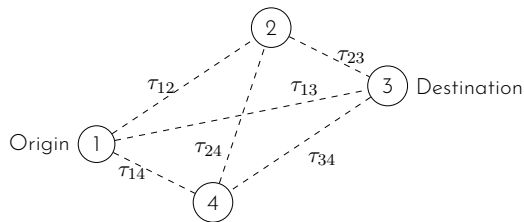


Taking a simple problem with a graph with 4 nodes.

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Taking a simple problem with a graph with 4 nodes.



The aim is to go from node 1 to node 3. τ_{ij} is the pheromone amount deposited on the edge (i, j) .

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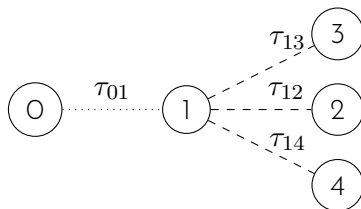
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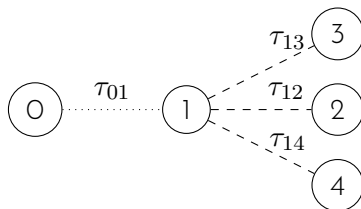
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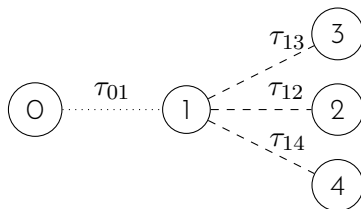
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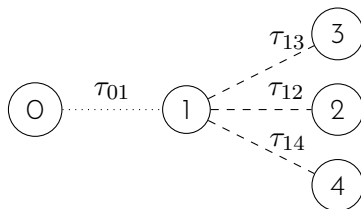
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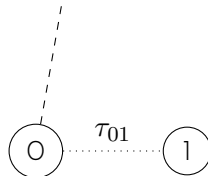


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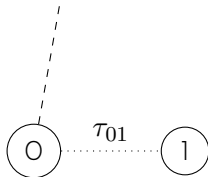
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In this case, we will consider edges (1,2), (1,3), and (1,4) only.



In cases where node 1 is not connected to any node other than node 0, we will then consider edge $(1, 0)$.



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This may create loops within the constructed path. These loops are removed once the path is fully constructed, i.e. once the destination node is reached.

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The path length L^k of ant k is calculated using the function chosen based on the questions. This is equivalent to the fitness function for GA and PSO.

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where ρ is a constant between 0 and 1 inclusively, which specifies the rate pheromones evaporate.

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The simplest form is $\Delta\tau_{ij}^k = \frac{1}{L^k}$.

The new pheromone concentration on the path, given that there are n_k ants in the system, is

$$\tau_{ij} = \tau_{ij} + \sum_{k=1}^{n_k} \Delta\tau_{ij}^k$$

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- ▶ terminate when an acceptable solution has been found
- ▶ terminate when all ants or most ants follow the same path

Algorithm 3 ACO algorithm

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ACO algorithm

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Select next node based on probability $p_{ij}^k(t)$ (Slide 37)

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repeat

Select next node based on probability $p_{ij}^k(t)$ (Slide 37)

Add edge (i, j) to path $x^k(t)$

until destination node is reached

end for

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end for

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end for

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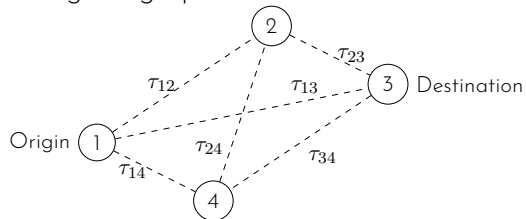
end for

 Update pheromone. $\tau_{ij}(t+1) = \tau_{ij}(t) + \sum_{k=1}^{n_k} \Delta\tau_{ij}^k(t)$

end for

Example

Taking the graph with 4 nodes.



Edge length is given by

Edge	Length	Edge	Length
(1,2)	3	(2,3)	10
(1,3)	20	(2,4)	1
(1,4)	5	(3,4)	3

Max-Min Ant System (MMAS)

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- ▶ ACO sometimes causes prematurely stagnation for complex problems, i.e. all ants follow exactly the same path with little exploration.
- ▶ MMAS restricts the pheromone intensities to within an interval. Different strategies have been developed to achieve this restriction of pheromone intensities.

Fast Ant System

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- ▶ A different pheromone update rule is used and no evaporation of pheromone happens.

Other swarm algorithms

- ▶ Ant colony clustering
- ▶ Bee algorithm
- ▶ Dragonfly algorithm
- ▶ Frogs algorithm
- ▶ Slime mold algorithm
- ▶ ...



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Thank you

CSC3034 Current Course Update

05 Swarm Intelligence - presentation

Synchronized from the current CSC3034 course site on 14 August 2026.

- Current Labs 4 and 5 cover PSO target search and ACO route finding.
- Physical examples account for robot speed limits, obstacles, and waypoint motion.
- Current runnable scripts are included in the synchronized Lab Solutions folders.

Current materials author: Dr Tang Tiong Yew

See the synchronized lab notes and runnable examples for full instructions.